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THE WEIGHTS OF THE BRAIN AND OF ITS PARTS, AND THE WEIGHT AND LENGTH OF THE SPINAL CORD IN THE DOG

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Many biometric studies of the human brain have been made, not only on individuals of various mental abilities but also of various races, and more recently the various divisions of the brain and its finer structure have received more attention quantitatively. Weights of the brain were compared to stature as early as 1836 by Parchappe. The brain of the albino rat has likewise been studied quantitatively very carefully in both its gross and microscopic aspects, but no attempt will be made to review the abundant literature in these fields. The brain of the dog has not received as much attention. There are several papers giving the weights of the entire dog brain (Hrdlicka, '07, Pezard, '36, and Crile and Quiring, '40). Kappers has studied the weights of the cortex ('26) and the cerebellum ('27) in various animals including the dog. The weights of the brain and of five of its divisions and of the cord for 11 dogs were included in a paper by Latimer ('24). Winterstein ('26) has gathered the available data on the water content and chemical analyses of various brains including the dog brain. There still seems to be the need of a statistical study on a more adequate number of dog brains and spinal cords.

The weights and the percentage weights of the brain and of seven of its divisions and the weight and length of the spinal cord from 321 dogs will be given. Then the variability of the various parts of the central nervous system will be presented and also the sex differences. Empirical formulae for determination of the weights of any of the divisions of the central nervous system from either body weight or body length will be presented as a more satisfactory method of estimating these weights than by the use of the averages or the percentages. The asymmetry of the two halves of the forebrain will be discussed, and then the correlations of the various parts of the central nervous system will be given.

MATERIALS AND METHODS

The brain, spinal cord, and the hypophysis were removed from 162 male and 159 female adult dogs. The dogs were those used for experiments in the physiology laboratory in which the animal was sacrificed after an experiment lasting only one laboratory period or less. In no case were animals used after prolonged experiments which would affect the general condition of the animal. All of the dogs were adults, as judged by the teeth and the ossification of the bones, and all were in good health. They were of all breeds and many of no recognizable breed, but they undoubtedly represent a good cross section of the animals used in routine laboratory experiments. Some came from the country and others from towns and cities. While the dogs were being used for this work all of the dogs from a certain laboratory section were taken unless they were immature or showed signs of disease. Toward the end only females were taken. The central nervous systems were taken from the dogs at various times from October, 1936, until July, 1939.

The dogs were weighed on a platform balance sensitive to ¼ lb. at the beginning of the experiment and these weights changed to the metric system were used, as it would have been impossible to have reweighed the animals after the experiment in which frequently more or less blood was lost. The dog was placed back up and the skin and most of the dorsal musculature was rapidly removed very soon after the killing of the animal. The head was severed from the vertebral column by an incision passing close to the occipital condyles which cut the cord from the medulla at approximately the level of the first root fila of the first spinal nerve. The laminae of the vertebrae were removed and the dura cut down the mid-dorsal line and turned back and while still attached to the vertebral column the length of the cord was measured from the cut edge to the beginning of the filum terminale by means of a flexible steel tape graduated to millimeters. Then the cord was removed and placed in a moist chamber together with the brain and hypophysis until taken to the laboratory and weighed.

The skin, muscles, the top of the skull, and the dorsal part of the dura were next removed and the brain allowed to fall gently into the palm of the hand from the inverted skull. The olfactory lobes were carefully separated from the cribiform plate and each pair of nerves were cut in order. The stalk of the hypophysis was cut as the brain was removed and later the hypophysis itself was carefully removed and its weight has been reported in a separate paper (Latimer, '41). The meninges, blood vessels, and nerves of some length were left on the brain and cord until they were prepared for weighing. Several medical students assisted in this part of the work, but the following final preparation of the parts for weighing and the separating of the brains into their several divisions as well as the weighing was all done by the author.

The dura and all large blood vessels were removed, all nerves were cut close to the cord, and then it was weighed in a glass stoppered weighing bottle on a chemical balance sensitive to a tenth of a milligram.

The brain was freed of all blood vessels, all nerves were cut close to the brain, and then it was divided in a manner similar to that employed in the study of the cat brain (Latimer, '38a and '38b). The pia mater is included in the brain weights, but all of the ventricles were emptied. The olfactory bulbs were separated from the rest of the brain at the very evident narrowing to form the olfactory tract. Next the cerebellum was removed by cutting the peduncles on a level with the margin of the walls of the medulla. This left the pons to be weighed with the medulla. The medulla and midbrain were separated at the isthmus, a very evident constriction. Then the separation of mesencephalon and diencephalon was made by an incision beginning dorsally on each side just anterior to the two anterior corpora quadrigemina and inferiorly cutting through just posterior to the stalk of the hypophysis. This incision was not a straight transverse cut but rather a very obtuse wedge shape with the mid portion the farthest anterior. The incisions were made part way through the brain stem beginning at both the upper and the ventral surfaces of this region and meeting in the middle. It was felt that this was a little more accurate method of making this, the most difficult separation of the parts of the brain. In nearly half of the dogs the remaining hemispheres and diencephalon were split into a right and a left half and these weighed separately. In these dogs the weight of the hemispheres and diencephalon is the sum of the weights of the two halves. The parts were put in glass stoppered weighing bottles as soon as dissected and weighed in the same manner as the cord. The total brain weight is the sum of the parts. It was felt that this would prevent additional handling of the brain and exposure to drying. All of the weights and measurements were made on the brain in the fresh condition.

The formulae used in the determination of the statistical constants are those usually so employed and are given by Dunn ('29). All of the computations were carefully checked, most of them by two workers.

First of all I wish to express my gratitude to Dr. O. O. Stoland

and to the other members of the Physiology Department for their kindness in permitting the use of this material, and especially to Mr. Fred Johnson who was so patient with us while we gathered this material. I also wish to express my appreciation of the careful assistance of the following medical students who helped with the removal of the brains and cords from the dogs; Mr. Edgar P. Sereres, Mr. Lelond Holbert, Mr. Milton H. Noltensmeyer, and Mr. Robt. R. Remsberg. To Miss Edna Mae McConnell and Mr. Douglas J. Malone I wish to express my appreciation of their careful, accurate work in assisting with the statistical part of this study.

Weights

The averages of the absolute and percentage weights and linear dimensions together with their coefficients of variation and the significant ratios are shown in Table 1. As has been stated above, this group of dogs is composed of all kinds and sizes, and with a range of body weight in the males from 4 to 30 kilograms and in the females, 4 to 24 kilograms, these averages are not very reliable, except of course for dogs of average size. Empirical formulae will be given later on for the more accurate estimation of the weights of the parts of the central nervous system. The data for the body weight and body length are given in the last two lines of Panel *A*.

The body weight is the most variable measurement in both sexes, followed by the weight of the spinal cord. The body length and the cord length, especially in the males, have very similar coefficients oi variation, which is to be expected. In the males, the weight of the entire brain is the least variable of all the ponderal measurements, while in the females there are three of the divisions of the brain slightly less variable. In the description of the methods of dissection of the brain parts it was suggested that the most difficult separation was between the relatively small mesencephalon and the much larger forebrain. Thus any irregularity in making this cut should be most evident in the weights of the midbrain. In the males, in Panel *A*, Table 1, the variability of the midbrain is fifth in order of increasing variability but still it is grouped with the parts of the hindbrain and just next in order above the medulla. In the females however it is the least variable of the seven parts of the brain. Hence it seems that the method of dividing the brain into its parts and especially this most difficult cut was done fairly well as shown by these coefficients of variability. The average of the first 10 coefficients of variation in

TABLE 1

MEASUREMENTS AND COEFFICIENTS OF VARIATION OF THE BRAIN AND OF ITS DIVISIONS, AND OF THE SPINAL CORD IN THE DOG

	162 male dogs		159 female dogs		Diff. / *PE* diff.
	Average weight or length	Coefficient of variation	Average weight or length	Coefficient of variation	
	A. *Weights and linear measurements*				
Brain	76.918 ± 0.580	14.24 ± 0.54	73.000 ± 0.570	14.60 ± 0.56	4.82
Olfactory bulbs	1.542 ± 0.017	21.41 ± 0.84	1.473 ± 0.018	23.31 ± 0.93	2.72
Hemispheres and dien.	61.975 ± 0.480	14.61 ± 0.56	58.967 ± 0.477	15.13 ± 0.59	4.44
Prosencephalon	63.517 ± 0.492	14.62 ± 0.56	60.438 ± 0.491	15.18 ± 0.59	4.43
Mesencephalon	2.045 ± 0.018	16.31 ± 0.63	1.915 ± 0.014	13.52 ± 0.52	5.78
Cerebellum	7.369 ± 0.061	15.59 ± 0.60	6.954 ± 0.053	14.20 ± 0.55	5.15
Medulla	3.986 ± 0.036	16.94 ± 0.65	3.691 ± 0.035	17.65 ± 0.69	5.91
Rhombencephalon	11.355 ± 0.091	15.12 ± 0.58	10.645 ± 0.083	14.51 ± 0.56	5.78
Cord weight	14.437 ± 0.181	23.63 ± 0.94	12.955 ± 0.180	25.95 ± 1.05	5.81
Cord length (cm.)	51.09 ± 0.37	13.41 ± 0.52	48.02 ± 0.40	15.33 ± 0.61	5.60
Body weight (kilos)	11.00 ± 0.26	44.75 ± 1.98	8.93 ± 0.22	46.13 ± 2.08	6.07
Body length (cm.)	81.9 ± 0.58	13.40 ± 0.51	76.8 ± 0.61	14.77 ± 0.57	6.03
	B. *Percentages of body weight*				
Brain	0.795 ± 0.013	31.76 ± 1.30	0.928 ± 0.015	30.23 ± 1.24	6.70
Olfactory bulbs	0.016 ± 0.00027	32.23 ± 1.33	0.018 ± 0.00028	28.33 ± 1.15	6.43
Hemispheres and dien.	0.640 ± 0.011	32.44 ± 1.34	0.751 ± 0.012	30.53 ± 1.26	6.82
Prosencephalon	0.656 ± 0.011	32.37 ± 1.33	0.769 ± 0.013	30.44 ± 1.25	6.93
Mesencephalon	0.021 ± 0.00037	33.09 ± 1.37	0.025 ± 0.00044	33.55 ± 1.40	5.91
Cerebellum	0.076 ± 0.0012	30.95 ± 1.27	0.089 ± 0.0015	32.00 ± 1.33	6.87
Medulla	0.041 ± 0.00063	29.33 ± 1.20	0.047 ± 0.00075	30.13 ± 1.24	6.13
Rhombencephalon	0.117 ± 0.0018	29.81 ± 1.21	0.135 ± 0.0023	31.10 ± 1.28	6.16
Cord weight	0.144 ± 0.0019	25.35 ± 1.01	0.160 ± 0.0022	26.00 ± 1.05	5.50
Cord length	0.527 ± 0.0085	29.62 ± 1.23	0.603 ± 0.0091	27.38 ± 1.09	6.10
	C. *Percentages of brain weight*				
Olfactory bulbs	2.00 ± 0.015	14.42 ± 0.55	2.01 ± 0.015	13.52 ± 0.52	0.47 (♀)
Hemispheres and dien.	80.54 ± 0.068	1.58 ± 0.059	80.72 ± 0.075	1.74 ± 0.066	1.74 (♀)
Prosencephalon	82.54 ± 0.064	1.46 ± 0.055	82.72 ± 0.071	1.61 ± 0.061	1.84 (♀)
Mesencephalon	2.67 ± 0.013	9.39 ± 0.35	2.64 ± 0.015	10.58 ± 0.40	1.76 (♂)
Cerebellum	9.60 ± 0.045	8.75 ± 0.33	9.57 ± 0.045	8.75 ± 0.33	0.49 (♂)
Medulla	5.19 ± 0.027	9.74 ± 0.37	5.07 ± 0.030	10.90 ± 0.42	2.98 (♂)
Rhombencephalon	14.78 ± 0.058	7.36 ± 0.28	14.63 ± 0.062	7.96 ± 0.30	1.77 (♂)

Panel *A* is 16.94 for the females and 16.59 for the males or, the females average a little over two per cent more variable than the males.

Panel *B* shows these weights and linear dimensions expressed as percentages of the body weight. The averages of these coefficients are 30.70 for the males, and 29.97 for the females, or the parts of the nervous system of the males are 1.85 times more variable when expressed as percentages of the body weight and for the females the ratio is 1.77 times. The absolute weights are slightly more variable in the females, but the percentages are more variable in the males by 2.4 per cent.

The last column gives the significant ratios, and all but one of the absolute weights and both of the linear dimensions are significantly greater in the males but all of the percentages of the body weight are significantly greater in the females. The only measurement in Panels *A* and *B* which does not show a significant difference is the weight in grams of the olfactory bulbs, and it is almost significant.

The weight of the cord is next to the body weight in variability in Panel *A*, but when the weights are reduced to percentages of the body weight, the lowest coefficient of variation is that of the cord in both sexes as shown in Panel *B*. In other words the weight of the cord tends to follow the variations in the body weight more than the weight of the brain or any of its parts. The medulla follows the body weight the best of any of the parts of the brain as shown by these coefficients in Panels *A* and *B*. The olfactory bulbs are the most variable of the parts of the brain in absolute weight in both sexes. A part of this may be due to the difficulty in removing all of the bulbs.

The weights of the parts of the brain expressed as percentages of the total brain weight are shown in Panel *C* of Table 1. Here we find the lowest coefficients of variation. The average for the males is 7.52 per cent and for the females, 7.88 per cent, or 4.8 per cent greater than for the males. It will be shown later that these percentages of the total brain weight do not change with the size of the dog and hence these percentages shown in Panel *C*, carry more weight than the percentage weights in Panel *B* or the absolute weights in Panel *A*. The last column again shows the ratios, none of which are significant although the medulla is almost significantly greater in the males. Although these are not significantly different, it is interesting to see that the first three, or the prosencephalon and its divisions, are greater in the females and the mesencephalon and the rhombencephalon are greater in the males. This posterior part of the brain is more intimately connected with the cord and also with the control of the body which of course is greater in the males, in size at least, if not in activity as well.

The earlier data on the weight of the dog nervous system (Latimer, '24) are not easily comparable with this series because the sexes are not separated and because of the small number of specimens. There is a much greater degree of variability in body weight and in the weights of all the parts of the nervous system in the smaller series. In this earlier series the brain stem (mesencephalon and medulla) seem to be a little lighter and they also form a somewhat smaller percentage of the total brain weight and the prosencephalon is heavier and forms a larger percentage of the brain weight, but in a series of only 11 doges these variations are not significant. Kappers ('26) gives the weight, but not the sex, of two dog brains as 78.5 and 77.7 grams. These are very close to the average brain weights in my dogs. Crile and Quiring ('40) give the total brain weight for ten male and eight female dogs. These weights expressed as percentages of the body weight are slightly lower than in my series and also they are slightly lower in the females than in the males.

In general, the weight of the brain and its parts, the weight and length of the cord and the body weight and length are with one exception significantly greater in the males. They average a little over two per cent more variable in the females. When these weights of the nervous system are expressed as percentages of the body weight, they are much more variable, with the greatest average variability in the males. All of the parts of the central nervous system, as percentages of the body weight, are significantly greater in the females. The parts of the brain expressed as percentages of the total brain weight are the most constant and there is no significant sex difference in their percentages although there is a suggested greater percentage in the weight of the forebrain and its parts in the females, and of the midbrain and the rhombencephalon in the males.

Empirical Formulae

The average weights and the average percentages given in Table 1 are obviously valid for dogs of approximately average body weight; but in any random selection of dogs, such as this series, a large proportion would be much above or below these averages and to better determine the weights of the various divisions of the central nervous system, empirical formulae have been developed by means of which the weight of any part of the central nervous system can be computed from either the body weight or the body length. This is the same method which was used for the central nervous system of the cat (Latimer, '40).

The weights of the brain and of its seven divisions, the weight and the length of the cord were each plotted against body weight and body length. The body weights were averaged for each two kilograms, the body lengths for each 10 centimeters increase in length, and the weights of the parts of the central nervous system were likewise averaged for each group and the common point for each two kilograms increase in body weight or for each 10 centimeters increase in body length were plotted and regression lines drawn through these points. Empirical formulae were developed for these regression lines. They were all straight lines and to save space they are not shown. The individual cases follow the lines with only a moderate amount of scatter.

There is a sex difference in the weights or length of all but one of the parts of the central nervous system and so the parts are plotted separately for the two sexes. The following formulae for the males, plotted on body weight, are valid between 4 and 26 kilos of body weight. In each of these formulae, Y represents the weight in grams or the length in centimeters of the part and X, the body weight in kilograms.

Brain,	$Y = 1.69X + 58.44$
Olfactory bulbs,	$Y = 0.046X + 1.066$ (4-18 Kilos, and from 18-26 K. average 1.93 grams)
Hemispheres and diencephalon,	$Y = 1.37X + 47.52$
Prosencephalon,	$Y = 1.37X + 49.32$
Mesencephalon,	$Y = 0.037X + 1.612$
Cerebellum,	$Y = 0.158X + 5.612$
Medulla,	$Y = 0.124X + 2.584$
Rhombencephalon,	$Y = 0.26X + 8.46$
Cord weight,	$Y = 0.61X + 7.76$
Cord length,	$Y = 1.37X + 35.92$

Using the same symbols the formula for the body length plotted on body weight is as follows:

$$Y = 2.02X + 59.92 \text{ from 4 to 26 kilos of body weight.}$$

The average percentage deviations of the calculated from the observed values for each two kilograms increase in body weight were determined and for the males they are as follows: brain, 1.72; olfactory bulbs, 4.26; hemispheres and diencephalon, 1.97; prosencephalon, 1.91; mesencephalon, 1.32; cerebellum, 1.89; medulla, 2.37; rhombencephalon, 1.88; cord weight, 2.04; cord length, 1.84 and for the body length on body weight, 1.40 per cent.

Similar formulae for the weights of the parts of the central nervous system of the female dogs plotted on body weight from 4 to 21 kilograms of body weight are given below. Y again represents the weight or length of the part and X, the body weight in kilograms.

Brain,	$Y = 2.06X + 54.76$
Olfactory bulbs,	$Y = 0.074X + 0.824$
Hemispheres and diencephalon,	$Y = 1.52X + 45.92$
Prosencephalon,	$Y = 1.72X + 45.12$
Mesencephalon,	$Y = 0.045X + 1.53$
Cerebellum,	$Y = 0.176X + 5.346$
Medulla,	$Y = 0.117X + 2.662$
Rhombencephalon,	$Y = 0.3X + 8.04$
Cord weight,	$Y = 0.68X + 6.93$
Cord length,	$Y = 1.67X + 32.92$
Body length (cm.),	$Y = 2.7X + 52.45$

The average percentage deviation of the calculated from the observed averages for these formulae are as follows: brain, 2.48; olfactory bulbs, 6.44; hemispheres and diencephalon, 2.53; prosencephalon, 2.72; mesencephalon, 1.89; cerebellum, 1.95; medulla, 4.01; rhombencephalon, 2.14; cord weight, 4.46; cord length, 2.38 and body length 2.52 per cent.

The formulae for these same parts of the central nervous system plotted on body length in centimeters are given below. Again Y represents the weight in grams or length in centimeters, and X now represents the body length measured in centimeters. These formulae are valid from 65 to 120 centimeters of body length for the males.

Brain,	$Y = 0.79X + 12.65$
Olfactory bulbs,	$Y = 0.022X + 0.26$
Hemispheres and diencephalon,	$Y = 0.62X + 11.60$
Prosencephalon,	$Y = 0.62X + 13.40$
Mesencephalon,	$Y = 0.018X + 0.57$
Cerebellum,	$Y = 0.076X + 1.19$
Medulla,	$Y = 0.052X + 0.26$
Rhombencephalon,	$Y = 0.13X + 0.82$
Cord weight,	$Y = 0.27X - 7.35$
Cord length,	$Y = 0.621X$

The average percentage deviations of the calculated from the observed values for the 10 centimeter groups are as follows: brain, 2.89; olfactory bulbs, 3.31; hemispheres and diencephalon, 2.85; prosencephalon, 2.86; mesencephalon, 1.96; cerebellum, 2.92; medulla, 2.47; rhombencephalon, 2.55; cord weight, 3.76 and cord length, 2.59 per cent.

The formulae for the females plotted on body length in centimeters, from 55 to 110 centimeters of body length are given below. Again Y represents the weight in grams or the length in centimeters of the part and X now represents the body length measured in centimeters.

Brain,	$Y = 0.72X + 17.70$
Olfactory bulbs,	$Y = 0.022X - 0.21$
Hemispheres and diencephalon,	$Y = 0.59X + 13.85$
Prosencephalon,	$Y = 0.58X + 16.10$
Mesencephalon,	$Y = 0.016X + 0.71$
Cerebellum,	$Y = 0.063X + 2.16$
Medulla,	$Y = 0.042X + 0.48$
Rhombencephalon,	$Y = 0.099X + 3.096$
Cord weight,	$Y = 0.26X - 6.49$
Cord length,	$Y = 0.63X - 0.11$

The average percentage deviations for the above formulae are: brain, 2.14; olfactory bulbs, 1.97; hemispheres and diencephalon, 2.47; prosencephalon, 2.28; mesencephalon, 1.75; cerebellum, 197; medulla, 2.31; rhombencephalon, 1.54; cord weight, 2.67, and cord length, 1.74 per cent.

This attempt to relate brain weight and body size is not new for it has been done many times for the human brain and Dubois ('14) has attempted to relate the total brain size and body size in various animals, but so far as is known it has never been attempted before for the dog nor for the parts of the dog brain.

The percentage weights of the brain, of its divisions and of the weight and length of the cord were plotted on body weight for both sexes in the same manner as the absolute weights. The averages were determined as described above for each two kilograms of body weight and through these points smoothed curves were drawn. All of the curves for both the males and the females were similar and to save space only a part of the curves for the males are shown in Figure 1. The smaller divisions of the brain are not shown but they may be determined easily by using the percentages of the entire brain weight (Table 1, Panel C) and the brain weight may be checked from Figure 1. The curves for the females resemble those of the males although they drop a little more rapidly.

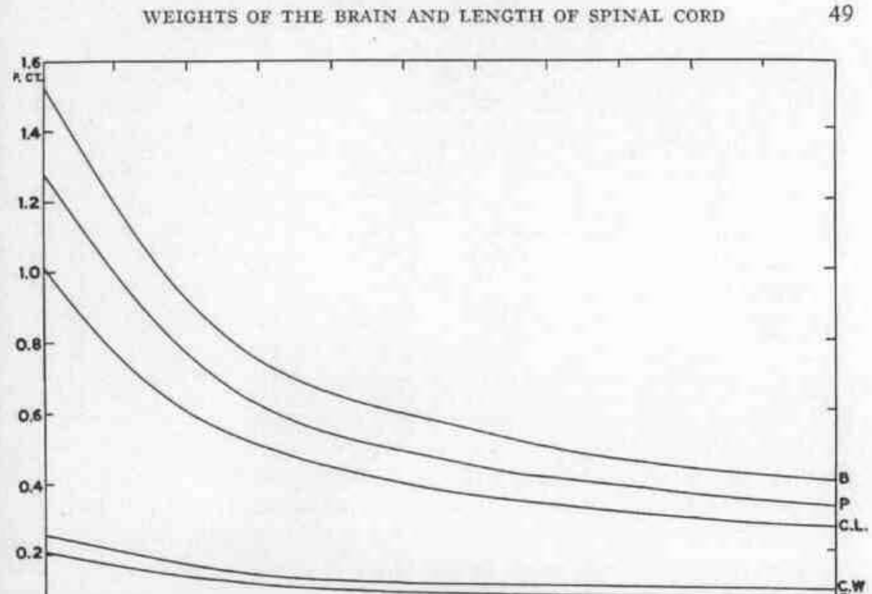

FIGURE 1

WEIGHTS OF SOME OF THE PARTS OF THE CENTRAL NERVOUS SYSTEM AS PERCENTAGES OF THE BODY WEIGHT AND PLOTTED ON BODY WEIGHT

The percentages are shown at the left and the body weights in kilograms at the bottom. The curves from above down are as follows; *B* represents the percentages of the total brain weight; *P* represents the percentages of the prosencephalon; *C.L.*, the cord length; *C.W.*, the cord weight; *R*, the rhombencephalon, and *M* represents the percentages of the mesencephalon.

The percentage values for each division of the central nervous system for both the males and the females for the smallest and the heaviest groups are given in Table 2 and the last column for each sex gives the percentage decrease. The smaller decreases are found in the cord and the hindbrain, parts which are more closely connected with the fundamental activities of the body. These percentage decreases are greater for the males in Table 2 but if we take into consideration the greater range of body weight, the percentage decrease per kilogram of body weight is greater in the females, or the curves decrease a little more rapidly in the females. The percentage decrease per kilo of increase in body weight varies for the different parts of course, but the average for the males is 3.4 per cent and for the females it is 4.0 per cent.

The weights of the seven divisions of the brain were reduced to

TABLE 2
CHANGES IN THE PERCENTAGES OF THE BODY WEIGHT

	Male dogs			Female dogs		
	Per cent at 5.08 K.	Per cent at 25.19 K.	Percentage decrease	Per cent at 5.19 K.	Per cent at 20.46 K.	Percentage decrease
Brain	1.334	0.405	69.64	1.250	0.459	63.28
Olfactory bulbs	0.0254	0.0078	69.29	0.0230	0.0104	54.78
Hemispheres and dien.	1.090	0.327	70.00	1.012	0.367	63.74
Prosencephalon	1.115	0.335	69.96	1.035	0.378	63.48
Mesencephalon	0.0354	0.0110	68.93	0.0345	0.0121	64.93
Cerebellum	0.1197	0.0375	68.67	0.1216	0.0425	65.05
Medulla	0.0634	0.0221	65.14	0.0633	0.0257	59.40
Rhombencephalon	0.1831	0.0596	67.45	0.1849	0.0682	63.12
Cord weight	0.2152	0.0916	57.43	0.204	0.102	49.72
Cord length	0.862	0.269	68.79	0.812	0.313	61.48

percentages of the total brain weight and the averages given in Panel C of Table 1. These individual percentages were plotted on body weight as was done for the percentages of the body weight and it was interesting to see that for each of the seven divisions of the brain for both the males and the females the resulting curves were horizontal straight lines, or the proportions of the brain are constant throughout the entire range of body weight in both sexes. Hence by using the percentages given in Panel C of Table 1, one can find the weight of any division of any sized dog brain whose entire weight is known and is within the given limits. Not only are these percentages constant throughout the entire range of body weight but there is little if any sex difference. The last column in Panel C, Table 1, gives the significant ratios and the only part which approaches a significant ratio is the ratio of 2.98 for the medulla. This is so nearly significant that we very likely should say that this part alone is significantly greater in its percentage of the total brain weight in the males. Kappers ('26) has determined the weight of the hemispheres in two dogs and he reports 86.8 and 83.3 per cent of the total brain weight. However he included what I have listed separately as midbrain and so with this added to my data, the males in my series would average 85.21 per cent and the females 85.36 per cent and the average of his two cases is 85.05 which means that his two cases fall within the normal range of variation of my series.

In general the parts of the central nervous system in both the males and in the females are relatively heaviest in the smallest dogs. The cord and the hindbrain decrease less in percentage of body weight than the other parts of the brain. The relative proportions of the

brains in both sex
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average weights of
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for the right half,
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hemisphere greater

brains in both sexes are remarkably constant throughout the entire range of body weight. This is shown by the low coefficients of variation shown in Panel *C* of Table 1, and also by the straight horizontal lines when these percentages of the brain weight are plotted on body weight. These percentages in Panel *C* are valid, within limits of individual variation of course, for the entire range of body weight.

Asymmetry

The asymmetry of many parts of the human body including the brain, has been carefully investigated and the question arose as to the symmetry of the forebrain in the dog. The olfactory bulbs were removed first and then, following the usual method of dissection outlined above, the hemispheres and diencephalon were carefully divided into two halves. The incision was carefully made from above and checked as it came through to the ventral surface of the brain and in all cases it appeared to be in the midplane as nearly as could be determined. This was done for 84 male dogs and 101 females. Each half of the forebrain, without the olfactory bulbs, was carefully weighed in a glass stoppered weighing bottle and the weight of each half recorded. For these specimens, the sum of the two halves was taken as the weight of the hemispheres and diencephalon.

Of the 84 male dogs, 46 or 54.76 per cent had the left half of the forebrain heavier, and for the females, 60 or 59.41 per cent were likewise heavier on the left side. Both of these percentags were so close to 50 per cent that it was decided to get the differences in the average weights of the two sides, and the average weight of the left half of the forebrain for the males was found to be 31.648 grams and for the right half, 31.632 grams, or the average weight of the left half is greater by 0.05 per cent. The similar averages for the females are, 29.87 grams for the left halves and 29.78 grams for the right halves. This means that in the female dogs the left half averages greater than the right by 0.30 per cent.

With these very slight differences in the average weights and with the frequency of a heavier left half in 55 and 59 per cent of the cases, it seems that we are justified in saying that there is no significant variation between the left and the right sides in the forebrain of the dog. Coghill ('36) finds a greater asymmetry of the ganglia in the young of the Amphibia. There seems to be no general agreement with regard to the human brain, for Andreassi ('33) finds the left hemisphere greater by 0.17 per cent in men and by 2.81 per cent in women. Higeta ('40) thinks that there is a greater difference in symmetry in the fetus than in the adult human brain and he lists five workers who find the left side greater and two who report a greater weight on the right side.

Correlations

The 10 parts of the central nervous system correlated with body weight, body length and also with brain weight are given in Table 3. The correlations between body weight and body length are given at the end of Panel *A*. All of the correlations are positive and all but one of the correlations in this table are above 0.5 per cent, and the aver-

TABLE 3

Correlations, Central Nervous System of the Dog

	162 male dogs	159 female dogs
A. *Correlations with body weight*		
Brain	+ 0.757 ± 0.023	+ 0.722 ± 0.026
Olfactory bulbs	+ 0.617 ± 0.033	+ 0.698 ± 0.027
Hemispheres and diencephalon	+ 0.746 ± 0.023	+ 0.688 ± 0.028
Proscencephalon	+ 0.731 ± 0.025	+ 0.697 ± 0.027
Mesencephalon	+ 0.542 ± 0.037	+ 0.678 ± 0.029
Cerebellum	+ 0.683 ± 0.028	+ 0.642 ± 0.031
Medulla	+ 0.795 ± 0.020	+ 0.772 ± 0.022
Rhombencephalon	+ 0.764 ± 0.022	+ 0.685 ± 0.028
Cord weight	+ 0.879 ± 0.018	+ 0.846 ± 0.015
Cord length	+ 0.435 ± 0.044	+ 0.940 ± 0.0064
Body length	+ 0.868 ± 0.013	+ 0.908 ± 0.0094
B. *Correlations with body length*		
Brain	+ 0.728 ± 0.025	+ 0.761 ± 0.023
Olfactory bulbs	+ 0.642 ± 0.031	+ 0.713 ± 0.026
Hemispheres and diencephalon	+ 0.704 ± 0.027	+ 0.714 ± 0.026
Proscencephalon	+ 0.714 ± 0.026	+ 0.735 ± 0.025
Mesencephalon	+ 0.538 ± 0.038	+ 0.723 ± 0.026
Cerebellum	+ 0.658 ± 0.030	+ 0.691 ± 0.028
Medulla	+ 0.740 ± 0.024	+ 0.775 ± 0.021
Rhombencephalon	+ 0.734 ± 0.024	+ 0.769 ± 0.022
Cord weight	+ 0.829 ± 0.017	+ 0.881 ± 0.012
Cord length	+ 0.875 ± 0.013	+ 0.981 ± 0.0021
C. *Correlations with brain weight*		
Olfactory bulbs	+ 0.750 ± 0.023	+ 0.765 ± 0.022
Hemispheres and diencephalon	+ 0.978 ± 0.0023	+ 0.991 ± 0.00098
Proscencephalon	+ 0.987 ± 0.0013	+ 0.994 ± 0.00069
Mesencephalon	+ 0.606 ± 0.034	+ 0.748 ± 0.024
Cerebellum	+ 0.833 ± 0.016	+ 0.807 ± 0.019
Medulla	+ 0.809 ± 0.018	+ 0.751 ± 0.023
Rhombencephalon	+ 0.869 ± 0.013	+ 0.805 ± 0.019
Cord weight	+ 0.777 ± 0.021	+ 0.744 ± 0.024
Cord length	+ 0.692 ± 0.028	+ 0.808 ± 0.019

age of the correlations in each of the three panels is greater in the females. The averages are greater in both sexes for the correlations with the body length and, as would be expected, still greater with the brain weight.

The length and the weight of the cord and the medulla form the three highest correlations with body weight and body length for both sexes, except for the cord length and body weight which is the lowest correlation in the males in Panel *A*. With this exception of cord length and body weight the order of the correlations is the same for the males, for both body weight and body length, or these correlations in decreasing order are: cord, medulla, rhombencephalon, total brain weight, prosencephalon, cerebellum, olfactory lobes, and mesencephalon. The high correlations of the hindbrain and the body dimensions are to be expected, but the cerebellum should be better correlated than it is if the cerebellum and the musculature, which forms a very large part of the total body weight, are as well related quantitatively as they are physiologically. The mesencephalon is largely a connecting pathway between the forebrain and the hindbrain and as shown in the earlier paper (Latimer, '24) it forms a much smaller part relatively, in the dog brain than it does in the anuran brain. The prosencephalon is sixth in order of the correlations in all four groups, males and females in Panels *A* and *B*.

Some of the correlations with the entire brain weight (Panel *C*) are high. The two highest, the prosencephalon and the hemispheres and diencephalon are very high but it must be remembered that they form approximately 83 and 81 per cent respectively of the total brain weight. If the order of decreasing correlations is checked with the percentage weights in Panel *C* of Table 1 it will be evident that these correlations are very nearly in the same order as the percentage weights of the total brain weight, or the relatively larger parts of the brain have the higher correlations with the total brain weight.

In general, the parts of the nervous system having the more primitive or fundamental relationships with the mass of the body, such as the cord and the medulla are better correlated with body weight and body length and the parts containing the higher associative centers are less well correlated with body size. Whether there is any correlation between the relative size of the prosencephalon and the "intelligence" of the dog we have no evidence, but the relative proportions of the brain do change without doubt from lower types of vertebrates to the more highly organized and in this larger sense there is a quantitative relationship between "intelligence" and relative proportions of the divisions of the brain.

TABLE 4

Correlations of the Parts of the Central Nervous System

	162 male dogs	159 female dogs
Olfactory bulbs and:		
Hemispheres and diencephalon	+ 0.669 ± 0.029	+ 0.731 ± 0.025
Prosencephalon	+ 0.708 ± 0.026	+ 0.746 ± 0.024
Mesencephalon	+ 0.427 ± 0.043	+ 0.609 ± 0.034
Cerebellum	+ 0.648 ± 0.031	+ 0.698 ± 0.027
Medulla	+ 0.652 ± 0.030	+ 0.693 ± 0.027
Rhombencephalon	+ 0.678 ± 0.029	+ 0.702 ± 0.027
Cord weight	+ 0.671 ± 0.029	+ 0.711 ± 0.027
Cord length	+ 0.642 ± 0.032	+ 0.785 ± 0.021
Hemispheres and diencephalon and:		
Prosencephalon	+ 0.988 ± 0.0013	+ 0.996 ± 0.00048
Mesencephalon	+ 0.578 ± 0.035	+ 0.725 ± 0.025
Cerebellum	+ 0.779 ± 0.021	+ 0.751 ± 0.023
Medulla	+ 0.766 ± 0.022	+ 0.689 ± 0.028
Rhombencephalon	+ 0.816 ± 0.018	+ 0.745 ± 0.024
Cord weight	+ 0.741 ± 0.024	+ 0.686 ± 0.028
Cord length	+ 0.661 ± 0.030	+ 0.781 ± 0.021
Prosencephalon and:		
Mesencephalon	+ 0.576 ± 0.035	+ 0.729 ± 0.025
Cerebellum	+ 0.788 ± 0.020	+ 0.769 ± 0.022
Medulla	+ 0.767 ± 0.022	+ 0.697 ± 0.028
Rhombencephalon	+ 0.809 ± 0.018	+ 0.752 ± 0.023
Cord weight	+ 0.728 ± 0.025	+ 0.715 ± 0.026
Cord length	+ 0.666 ± 0.030	+ 0.784 ± 0.021
Mesencephalon and:		
Cerebellum	+ 0.534 ± 0.038	+ 0.685 ± 0.028
Medulla	+ 0.668 ± 0.029	+ 0.872 ± 0.013
Rhombencephalon	+ 0.611 ± 0.033	+ 0.786 ± 0.020
Cord weight	+ 0.645 ± 0.031	+ 0.853 ± 0.015
Cord length	+ 0.543 ± 0.038	+ 0.738 ± 0.025
Cerebellum and:		
Medulla	+ 0.763 ± 0.022	+ 0.765 ± 0.022
Rhombencephalon	+ 0.962 ± 0.0039	+ 0.914 ± 0.0088
Cord weight	+ 0.725 ± 0.025	+ 0.775 ± 0.021
Cord length	+ 0.661 ± 0.031	+ 0.724 ± 0.026
Medulla and:		
Rhombencephalon	+ 0.889 ± 0.011	+ 0.845 ± 0.015
Cord weight	+ 0.906 ± 0.0095	+ 0.898 ± 0.010
Cord length	+ 0.745 ± 0.024	+ 0.801 ± 0.020
Rhombencephalon and:		
Cord weight	+ 0.852 ± 0.015	+ 0.808 ± 0.019
Cord length	+ 0.738 ± 0.025	+ 0.773 ± 0.022
Cord weight and:		
Cord length	+ 0.890 ± 0.011	+ 0.937 ± 0.0067

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$+0.711 \pm 0.027$
$+0.785 \pm 0.021$
$+0.996 \pm 0.00048$
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$+0.751 \pm 0.023$
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$+0.745 \pm 0.024$
$+0.686 \pm 0.028$
$+0.781 \pm 0.021$
$+0.729 \pm 0.025$
$+0.769 \pm 0.022$
$+0.697 \pm 0.028$
$+0.752 \pm 0.023$
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$+0.808 \pm 0.019$
$+0.773 \pm 0.022$
$+0.937 \pm 0.0067$

The correlations between each of the seven parts of the brain and the weight and length of the cord and the other eight divisions are shown in Table 4. All of these are positive and all but one are significant. All of the female correlations are 0.6 or above. In the males, there is one correlation of 0.4 (olf. bulbs and midbrain) and four between 0.5 and 0.6 and these are correlations of the mesencephalon with the prosencephalon, hemispheres and diencephalon, cerebellum and cord length. The mesencephalon seems to have the lowest correlations with the other parts of the brain and spinal cord. Of the 36 pairs of correlations, 12 are greater in the males and 24, in the females or in general the parts of the central nervous system are better correlated in the female dogs. The parts which are better correlated in the males than in the females are the rhombencephalon and its two subdivisions correlated with the forebrain and the hemispheres and diencephalon, the rhombencephalon with its two divisions and the cord correlated with the medulla and the rhombencephalon. The highest correlations are between the forebrain and the hemispheres and diencephalon but this is to be expected when we correlate any structure with such a large subdivision of the total. The correlations between the rhombencephalon and its two divisions are also high but not as great as that between the forebrain and hemispheres. The correlations between the weight and length of the cord are high, especially in the female.

The correlations in this table as well as the constancy of the percentages of the parts of the brain with reference to the entire brain weight, show that the proportions of the brain are uniform and do not change with the size of the dog.

Summary

The averages, the coefficients of variation, and the significant ratios are given for the weights in grams of the male and the female brains, for the seven divisions of the brain, for the weight and length of the cord and for the body weight and body length. Similar statistical constants are given for the weights of these divisions of the central nervous system as percentages of the body weight and also for the seven divisions of the brain as percentages of the total brain weight.

The body weight is the most variable followed by the cord weight. The male brain is the least variable in weight but in the females there are three divisions of the brain which are less variable than the total brain weight. The coefficients of variation are much larger for these weights when given as percentages of the body weight. The least variable are the seven parts of the brain given as percentages of the total brain weight. In all of these groups the females are generally slightly more variable than the males.

The weights in grams, with one exception, are significantly greater in the males but all of the percentages of the body weight are significantly greater in the females. There is no significant difference in the seven parts of the brain given as percentages of the total brain weight, but the forebrain in the females tends to be larger and the midbrain and hindbrain are slightly larger in the males.

Empirical formulae are given as a more accurate method of estimating the weights of any of the parts of the central nervous system from either the body weight or the body length for both the males and for the female dogs. The curves for the parts of the nervous system as percentages of the total body weight are all concave superiority, with the larger percentages in the smaller dogs. The parts of the brain as percentages of the total brain weight are all horizontal straight lines, which means that these percentages of the total brain weight are valid throughout the entire range of body weight for both the male and the female dogs.

The correlations of the parts of the system average a little higher with the body length than with the body weight and the correlations between the entire brain weight and the weight of its seven divisions are naturally still greater. All of the correlations are positive and all but one are significant. As a rule they are somewhat higher in the female dogs. The correlations between the various divisions of the nervous system are likewise positive and good correlations. The mesencephalon has the lowest correlations with the body weight, with the body length, with the total brain weight and with the other divisions of the system. The spinal cord and the medulla have the highest correlations with the body weight and the body length. The prosencephalon is sixth in order of correlations or a little below the average in each group of correlations. This seems to show that the cord and the medulla are more closely related to the mass of the body, and the forebrain seems to show some degree of independence of body size as far as its weight is concerned.

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